

An on-chain venue trails the centralised market by half a second, in every hour measured

Thomas Erhel
Quasareum, Paris, France
`contact@quasareum.com`
ORCID 0009-0007-1772-9892

July 29, 2026

Abstract

Keywords: lead–lag; price discovery; market microstructure; decentralised exchange; perpetual futures; Hayashi–Yoshida.

JEL: G14, G15, C58, C81.

© 2026 Quasareum. Licensed under a Creative Commons Attribution 4.0 International License (CC BY 4.0), <https://creativecommons.org/licenses/by/4.0/>.

- 1 Introduction
- 2 Related work
 - 2.1 Lead–lag estimation between venues
 - 2.2 Centralised versus decentralised price discovery
 - 2.3 What is new here
- 3 Setting and data
 - 3.1 Hyperliquid and the comparison venues
 - 3.2 The tape
 - 3.3 Sample
- 4 The estimator, and what it was checked against
- 5 Result: a stable half-second lag
 - 5.1 Magnitude and universality
 - 5.2 The control that makes it Hyperliquid’s
 - 5.3 Volatility
- 6 What the lag is not
 - 6.1 Not an artefact of unequal observation rates
 - 6.2 Not the follower’s own price innovations
 - 6.3 Not endogenous sampling
 - 6.4 What that leaves
- 7 Two venues, two clocks
- 8 [TEST C] Quotes or prints?
- 9 Implications
- 10 Limitations
- 11 Reproducibility

Declarations